EAMHP

Excel-Based Avionics Mechanical & Health Prediction

SELF-DEBUGGING GUIDE

Issue Diagnosis & Resolution Reference — V1.1.0

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Issues Covered	5 Issue Categories | 4-Step Debug Protocol | Best Practices

1. Overview

This Self-Debugging Guide provides structured diagnostic and resolution procedures for all known operational issues within the EAMHP V1.1.0 system. It is intended to be used by AME students, avionics trainees, and technical operators whenever the system does not behave as expected.

🛠️ How to Use This Guide

Identify the symptom you are experiencing from the section headers below. Each issue section provides a two-column layout: Possible Causes on the left and numbered Resolution Steps on the right. Work through the resolution steps in order. If the issue persists after completing all steps, escalate to the 4-Step Debug Protocol in Section 3.

EAMHP issues typically fall into one of five categories:

Issue ID	Issue Category	Primary Symptom	Severity
ISS-01	No Output Generated	System produces no fault prediction or status output	HIGH — Check immediately
ISS-02	Incorrect Fault Prediction	Fault_ID or category does not match expected diagnosis	HIGH — Review inputs and logic
ISS-03	Excel Formula Errors	#VALUE!, #REF!, #N/A or other formula error in any cell	MEDIUM — Formula/reference issue
ISS-04	Macro Not Running	VBA macros silently fail or produce no result on trigger	HIGH — Check Excel security settings
ISS-05	Slow Performance	Workbook becomes unresponsive or evaluates slowly after data entry	LOW — Optimisation required

2. Common Issues & Resolution

Each issue below is presented with a full causes-and-fixes card. Work through the resolution steps in numbered order.

⛔ Issue ISS-01 — No Output Generated

⚠️ Possible Causes	✅ Resolution Steps
• VBA macros are disabled in Excel Trust Center settings • One or more required InputData fields are empty (Engine Type, Parameter, Value)	1. Open Excel → File → Options → Trust Center → Trust Center Settings → Macro Settings → Select 'Enable all macros' or 'Enable macros with notification', then close and reopen EAMHP

• Logic evaluation formula not triggered due to empty row or incorrect table reference • DataValidation sheet has been accidentally modified or deleted • File was opened in Protected View (read-only mode from email/download)	2. Ensure all required InputData fields are populated: Engine Type (dropdown), Parameter Name (dropdown), Unit (dropdown), and Value (numeric) 3. Check that the InputData row is within the defined Excel table range. If rows were added outside the table, extend the table boundary to include them 4. If DataValidation sheet is missing or corrupted, restore from the backup copy of EAMHP_V_1_0_0.xlsm 5. If opened from email or download, click 'Enable Editing' in the yellow Protected View banner at the top of the Excel window before using the tool

⚠️ Issue ISS-02 — Incorrect Fault Prediction

⚠️ Possible Causes	✅ Resolution Steps
• Parameter value entered for the wrong engine type (e.g., CFM56-5B value entered with V2500 selected) • Reference tolerance values in ReferenceData sheet have been accidentally overwritten or modified • Two or more parameters are faulted but only one is being evaluated (incomplete input session) • Logic overlap between fault categories causing incorrect category assignment • Fault_ID in FaultMatrixEngine sheet does not match the expected combination mapping	6. Recheck all InputData entries: confirm Engine Type matches the actual aircraft engine variant being evaluated 7. Cross-verify the Upper and Lower Tolerance values auto-populated in InputData against the certified AMM limits for the parameter and engine type. If values differ, the ReferenceData sheet may be corrupted — restore from backup 8. Ensure all faulted parameters from the same maintenance session are entered as separate rows in InputData before querying fault output 9. Review the Fault_Category assigned in the output against the FaultMatrixEngine sheet. If the category appears incorrect, check for duplicate or overlapping fault ID entries in that sheet 10. If issue persists, cross-reference the fault combination manually against EAMHP_fault_matrix_with_engine_procedures.xlsx to validate the expected Fault_ID and category

📊 Issue ISS-03 — Excel Formula Errors (#VALUE!, #REF!, #N/A)

⚠️ Possible Causes

- #N/A in Fault Status: Parameter Name or Engine Type entry does not match any record in ReferenceData
- #REF! in Tolerance columns: A row or column that a formula references has been deleted or moved
- #VALUE! in Fault Status: Non-numeric data entered in the Value field (e.g., text, spaces, or special characters)
- Protected sheet modification: A formula in ReferenceData or FaultMatrixEngine has been accidentally overwritten
- Excel version incompatibility causing formula evaluation differences (rare, Excel 2016 vs 365)

✅ Resolution Steps

11. For #N/A: Verify that the Parameter Name and Engine Type are selected from the dropdowns exactly as they appear in the DataValidation sheet. Free-text entry with slightly different spelling will cause a lookup failure
12. For #REF!: Press Ctrl+Z to undo recent changes. If the error persists, use the formula auditing tool (Formulas ribbon → Trace Precedents) to identify the broken reference and restore the original formula from backup
13. For #VALUE!: Clear the Value field and re-enter a numeric value only. Do not type units or text in the Value column — units are handled by the separate Unit dropdown
14. For protected sheet edits: Close the workbook without saving and reopen from the backup copy to restore formula integrity
15. For Excel version issues: Ensure the workbook is saved in .xlsm format. Avoid saving as .xlsx (strips macros) or older .xls format

⚙️ Issue ISS-04 — Macro Not Running

⚠️ Possible Causes

- Excel macro security settings are set to 'Disable all macros without notification'
- EAMHP was opened from an untrusted location (network drive, email attachment, USB) without adding it to Trusted Locations
- VBA project code has been corrupted due to improper file save (e.g., saved as .xlsx instead of .xlsm)
- Excel add-in conflict or another macro-enabled workbook interfering with EAMHP VBA execution

✅ Resolution Steps

16. Go to File → Options → Trust Center → Trust Center Settings → Macro Settings and select 'Disable all macros with notification', then close and reopen EAMHP. Click 'Enable Content' when prompted
17. Add the EAMHP file location to Trusted Locations: Trust Center Settings → Trusted Locations → Add new location → browse to EAMHP folder
18. Verify the file is saved with .xlsm extension. Open the VBA editor (Alt+F11) to confirm macro modules are present and not empty

• VBA runtime error halted execution mid-run and was not cleared before next trigger attempt	19. Close all other open Excel workbooks and retest EAMHP macros in isolation to rule out add-in conflicts 20. Open VBA editor (Alt+F11), check for any highlighted error lines in the code and press Reset (Run menu → Reset) to clear the halted state, then retry the macro trigger

⏱️ Issue ISS-05 — Slow Performance

⚠️ Possible Causes	✅ Resolution Steps
• Automatic calculation is recomputing all formulas across the entire workbook on every cell change • Large number of InputData rows causing VLOOKUP recalculations across the full dataset • Multiple VBA macros triggering simultaneously via Worksheet_Change events on each keystroke • Conditional formatting applied to excessively large ranges (e.g., entire columns instead of defined table rows) • Workbook opened on low-specification hardware with insufficient RAM for large formula arrays	21. Temporarily switch Excel to manual calculation during bulk data entry: Formulas ribbon → Calculation Options → Manual. Press F9 to recalculate when ready to evaluate 22. Limit InputData entries to the current maintenance session only. Archive completed sessions by copying data to Sheet2 (reserved) and clearing InputData rows before starting a new session 23. Review VBA code in the editor (Alt+F11) and check if Worksheet_Change events are firing on every cell. Add Application.EnableEvents = False at the start of macros and re-enable at the end to prevent cascading triggers 24. Review and reduce conditional formatting scope: select the relevant column, open Conditional Formatting → Manage Rules, and restrict rules to the actual data range rather than entire columns 25. Close other open applications to free available RAM. EAMHP V1.1.0 is designed to run within 200 MB RAM and should be responsive on any modern office machine

3. 4-Step Debugging Protocol

If the individual issue cards in Section 2 do not resolve your problem, follow this structured 4-step debugging protocol. Work through each step in order before escalating.

🧠 When to Use This Protocol

Use the 4-step protocol when: (1) you cannot identify which Issue ID matches your symptom, (2) multiple issues are occurring simultaneously, (3) a resolution step in Section 2 has introduced a new problem, or (4) the workbook behaves differently after a recent edit.

STEP 1

Verify Input Data Integrity

Before investigating logic or formulas, confirm that the data entering the system is correct. The majority of EAMHP issues originate from invalid or incomplete input data.

- Confirm Engine Type is selected from the dropdown (not free-typed)
- Confirm Parameter Name matches exactly one of the 16 valid EPP names
- Confirm Value field contains a numeric value only — no units, text, or spaces
- Confirm Unit is selected from the dropdown
- Confirm no InputData rows are partially filled (all required fields populated or row is empty)

STEP 2

Validate Parameter Ranges Against Reference

If input data is correct but Fault Status appears wrong, manually verify the tolerance values being used for evaluation.

- Navigate to the ReferenceData sheet and locate the row matching your Parameter Name and Engine Type
- Compare the Lower Tolerance and Upper Tolerance values against the current Airbus A320 AMM for the relevant engine
- Verify the auto-populated Upper Tolerance and Lower Tolerance in InputData match the ReferenceData values
- If tolerance values in ReferenceData appear incorrect or blank, restore from backup copy
- Check that the Parameter Name in InputData uses the exact same string as the ReferenceData Parameter Name column (case-sensitive VLOOKUP)

STEP 3

Trace Logic Step by Step

Manually trace the evaluation path for the faulted parameter through each logic stage to identify where the breakdown occurs.

- Stage 1 — Is Fault Status = FAULT or NORMAL? If #N/A, fix the input lookup issue first

- Stage 2 — Is the state label (HIGH/LOW) consistent with the value vs. tolerance comparison?
- Stage 3 — Open FaultMatrixEngine sheet: does the faulted parameter combination exist as a Fault_ID row?
- Stage 4 — Open the Mixed Report sheet: does the identified Fault_ID have populated Corrective_Action, Preventive_Action, CFM56-5B Procedure, and V2500 Procedure columns?
- Stage 5 — Is the output displaying in the correct category sheet for the Fault_Category assigned?

STEP 4

Verify Output Mapping

Confirm that the maintenance output displayed to the user accurately reflects the fault matrix data for the identified Fault_ID.

- Cross-reference the displayed Fault_ID against EAMHP_fault_matrix_with_engine_procedures.xlsx to verify the fault record
- Confirm that the CFM56-5B and V2500 procedures shown match the engine variant selected in InputData
- If output fields are blank: check for null values in the Mixed Report sheet for that Fault_ID
- If output is showing a different Fault_ID than expected: check for duplicate parameter combinations in FaultMatrixEngine — this indicates a data integrity issue requiring backup restoration
- Document the Fault_ID, input parameters, and unexpected behaviour for reporting to the EAMHP developer

4. Quick Reference — Error Code Diagnostic Table

Use this table for rapid identification of Excel error codes that may appear in the EAMHP InputData or output sheets.

Error Code	Where It Appears	Most Likely Cause in EAMHP	Immediate Action
#N/A	Fault Status, Upper/Lower Tolerance	Parameter Name or Engine Type doesn't match any ReferenceData record	Reselect Parameter and Engine Type from dropdowns
#VALUE!	Fault Status column	Non-numeric data entered in Value field	Clear Value field and enter numeric reading only
#REF!	Tolerance columns, output fields	A referenced row/column has been deleted or the formula range was broken	Press Ctrl+Z to undo, or restore from backup
#NAME?	Any formula cell	A named range referenced in a formula has been deleted or renamed	Open Name Manager (Formulas → Name Manager) and restore missing names

Error Code	Where It Appears	Most Likely Cause in EAMHP	Immediate Action
#DIV/0!	Any calculated field	A formula is dividing by a cell that is empty or zero	Check for empty denominator cells in formula logic
#NULL!	Any formula cell	Incorrect range operator used in a formula (rare in EAMHP)	Check formula syntax in the affected cell
Circular	Status bar warning	A formula refers back to its own cell, creating a circular dependency	Formulas ribbon → Error Checking → Circular References to locate and fix

5. Best Practices for EAMHP Operation

Following these best practices will prevent the majority of issues described in this guide and ensure EAMHP operates reliably across all maintenance sessions.

	Best Practice	Details and Guidance
💾	**Always Keep a Backup**	Before making any changes to the EAMHP workbook — including adding data, modifying reference values, or editing VBA — save a dated backup copy (e.g., EAMHP_V_1_0_0_backup_YYYYMMDD.xlsm). This is your recovery option for all corruption and formula issues.
✅	**Use Dropdowns Only**	Always select Engine Type, Parameter Name, and Unit from the validated dropdowns. Never free-type these fields. Even a single character difference (space, capitalisation, or abbreviation) will cause VLOOKUP to return #N/A and prevent fault evaluation.
🔒	**Never Modify Protected Sheets**	ReferenceData, FaultMatrixEngine, and DataValidation sheets contain certified reference data. Modifying these sheets will corrupt the logic engine and produce incorrect outputs. If sheet protection is off, treat these sheets as read-only.
🧹	**Clear InputData Between Sessions**	When starting a new maintenance session, clear the previous InputData entries. Carrying over old data from a different engine or parameter set will corrupt multi-parameter fault correlation and produce incorrect Fault_IDs.
📋	**One Engine Type Per Session**	All InputData rows within a single fault evaluation session must use the same Engine Type. Mixing CFM56-5B and V2500 entries in the same session will cause cross-variant tolerance mismatches and invalid fault combinations.
🔄	**Keep Excel Updated**	EAMHP is developed and tested on Microsoft Excel 365. Older Excel versions (pre-2016) may not support all formula types used. Keep your Excel installation updated to ensure full formula and VBA compatibility.
⚠️	**EAMHP is a Support Tool Only**	EAMHP outputs are for maintenance planning support and training purposes only. All maintenance actions must be cross-verified against the current issue of the Airbus A320 AMM and TSM before execution. EAMHP does not replace the judgement of a Type Rated, licensed AME.

	Best Practice	Details and Guidance
📝	**Document Unusual Behaviour**	If EAMHP produces unexpected outputs that cannot be resolved using this guide, document the exact inputs, the unexpected output, and the steps already taken. Report to the EAMHP developer (Nikhilesh Avionics Projects) via GitHub for issue tracking.